

Part 1 | Project Proposal

Predicting student dropout and academic success

CS-13410 Introduction to Machine Learning · Fall 2026 · Assignment 1

Problem and target

Predict a university student's eventual **Dropout, Enrolled, or Graduate** outcome using data known by the end of the second semester. This is a three-class classification problem. The target is the CSV column **Target**. The semester outcomes are available at the proposed prediction time; this is not an enrollment-time forecast.

Dataset and suitability

The supplied data.csv matches the UCI “Predict Students' Dropout and Academic Success” dataset: 4,424 students, 36 predictors, and one target. It combines academic counts/grades, demographics, application categories, and economic indicators. Its size and mixed numeric/categorical features support the required preprocessing and later model comparisons. Classes: Graduate 2,209; Dropout 1,421; Enrolled 794. There are no missing cells or exact duplicate rows in this supplied copy. UCI lists the dataset under CC BY 4.0.

Success measure

Use **macro F1** as the primary metric so all three outcomes receive equal weight. Later model evaluations will also report per-class recall, especially for the smaller Enrolled class. A stratified 80/20 train/test split (random state 42) reserves the test partition for final evaluation; cross-validation in later parts will occur on training data.

Part 1 preparation plan

Create first- and second-semester approval rates. Median-impute continuous features, cap numeric outliers using train-fitted IQR bounds, and robust-scale them. Most-frequent-impute categorical fields and one-hot encode them; preserve binary flags. All operations are inside a scikit-learn Pipeline/ColumnTransformer fitted to training data only. No global resampling is done. Part 2 will address imbalance with stratified folds and model-level weighting where appropriate.

Limitations and next steps

Semester-2 records can only support prediction after that semester. Academic measurements may strongly reflect outcomes already taking shape, so deployment timing must be explicit. The data originate in Portuguese higher education; performance and fairness may not transfer to other institutions. Instructor dataset approval and group registration remain for the team to complete. Parts 2–4 will extend the same notebook with models and evaluation.

Source: Realinho et al. (2021), UCI Machine Learning Repository, doi:10.24432/C5MC89. License: CC BY 4.0.